Weyland-Yutani mine output analytics

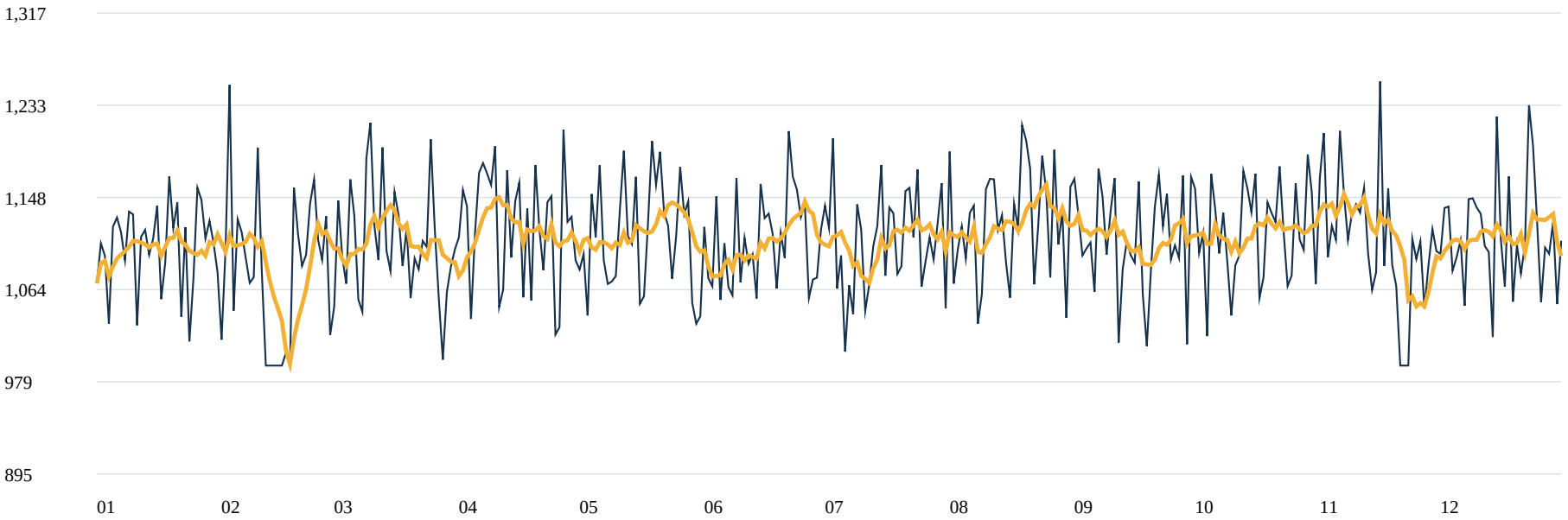
Formula-driven synthetic production, configurable scenarios, anomaly detection and operational trend review

5	365	6	4
active mines	daily periods	event scenarios	anomaly methods

This report accompanies an editable workbook that generates one year of daily production for five mines. It combines configurable distributions, calendar factors, trend, smoothing and six operational events with a dashboard and four independent anomaly checks.

Acheron Prime - synthetic 2026 scenario

2026-01-01 to 2026-12-31 | Tonnes



Month Blue: daily output Gold: trailing 7-day average

Model design and controls

The workbook keeps editable assumptions separate from generated data and analysis. Yellow cells are intended for user input; output, chart, and statistical cells are formula-driven.

Capability	Implementation
Distribution	Uniform bounds plus Box-Muller normal variation using RAND(), mean and standard deviation.
Smoothing	Editable alpha blends each random observation toward the mine's configured operating mean.
Calendar effects	Monday-Sunday factors alter output while preserving the underlying mine distribution.
Events	Six mine-specific date windows add positive or negative output impacts.
Trend	Per-mine monthly trend factors accumulate across the selected date range.
Anomalies	IQR fences, absolute z-score, distance from seven-day moving average, and a Grubbs G comparison.
Charts	Daily line/moving-average chart, mine comparison bars, stacked anomaly counts, and a polynomial chart trendline.

Editable assumptions

Mining output generator assumptions

Global setting	Editable value
Start date	2026-01-01
End date	2026-12-31
Mine count	5
Selected mine	Acheron Prime
Chart type	Line
Trendline degree	2
IQR multiplier	1.5
Z-score threshold	3

Generator setting	Editable value
Smoothing alpha	0.35
Random seed note	Use Formulas > Calculate Now to regenerate RAND-based output

Mine	Active	Mean t/day	Std dev	Minimum	Maximum	Trend / month
Acheron Prime	1	1120	115	760	1540	0.20%
LV-426 North	1	930	95	620	1300	-0.10%
Nostromo Deep	1	1260	145	820	1740	0.30%
Hadley's Hope	1	845	80	590	1140	0.10%
Yutani Ridge	1	1050	125	700	1480	0.25%
Reserve 6	0	900	100	600	1300	0.00%
Reserve 7	0	900	100	600	1300	0.00%
Reserve 8	0	900	100	600	1300	0.00%

Weekday	Output factor
Monday	1.03
Tuesday	1.02
Wednesday	1.01
Thursday	1.00
Friday	0.98
Saturday	0.91
Sunday	0.87

Event	Mine	Start	End	Output impact
Scheduled overhaul	Acheron Prime	2026-02-12	2026-02-18	-35%
Ore seam boost	LV-426 North	2026-03-10	2026-03-25	18%
Power interruption	Nostromo Deep	2026-05-03	2026-05-06	-55%
Crew rotation	Hadley's Hope	2026-07-01	2026-07-09	-12%
Equipment upgrade	Yutani Ridge	2026-08-15	2026-09-10	22%
Storm shutdown	Acheron Prime	2026-11-21	2026-11-23	-80%

Interpretation note

The workbook is synthetic by design. Recalculation regenerates RAND-based observations, so exact values and anomaly counts change while the configured distributions, event effects and formula logic remain stable. The Grubbs critical value uses a transparent sample-size approximation for the 0.05 level to maintain compatibility across Excel and Google Sheets.